PHASE 2 SPECIFICATION

PlanSearch

National Expansion

All 31 Irish local authorities. Full project lifecycle from planning application to completed building. AI-generated construction value estimates and commercial significance scoring. The free, open alternative to CIS Ireland.

30+	362k	9	€12
COUNCILS VIA SINGLE NATIONAL API	APPLICATIONS ALREADY IN NPAD	LIFECYCLE STAGES TRACKED	MONTHLY INFRASTRUCTURE COST

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VERSION	DATE	BUILDS ON	DATA SOURCES
1.0	March 2026	Phase 1 Spec v1.1	NPAD · BCMS · CRO · DCC CSV

For Claude Code

PlanSearch — Phase 2 Technical Specification

National Expansion + AI Intelligence Layer + Post-Planning Lifecycle Tracking

Version: 1.0 | **Date:** March 2026 **Builds on:** PlanSearch Phase 1 (Dublin City Council MVP) **Outcome:** A free, national, AI-powered planning intelligence platform covering all 31 Irish local authorities — from planning application through to completed building — that outperforms CIS Ireland on every data dimension except human network intelligence.

Executive Summary

Phase 1 built a fast, beautiful, searchable database of Dublin City Council planning applications. Phase 2 does three transformative things:

1. Go national. The Department of Housing operates a single ArcGIS REST API — the National Planning Application Database (NPAD) — that already publishes 362,000+ planning applications from 30 of 31 local authorities under a CC BY 4.0 licence. Ingesting this makes national coverage an engineering task measured in days. The 31st council (Kerry) has its own APAS-based open data feed. The NPAD also already includes applicant name, floor area, residential unit count, land use code, and a direct link to each application's page on the local council portal — fields that CIS extracts with human researchers.

2. Add AI intelligence. Using structural data already present in the national dataset (floor area, unit count, site area, land use code, applicant name) combined with Claude, PlanSearch can automatically estimate construction values, score commercial significance, identify one-off houses vs apartment blocks vs commercial developments, and publish weekly digest alerts — replicating the core output of CIS's human research team at near-zero marginal cost.

3. Track the full project lifecycle. The National Building Control and Market Surveillance Office (NBC&MSO) publishes two free open datasets via data.nbco.gov.ie: Commencement Notices (filed when construction starts) and Certificates of Compliance on Completion (filed when a building is finished and certified). Both datasets include the planning permission reference number as a direct join key. Combined with the Fire Safety Certificate / Disability Access Certificate applications dataset, PlanSearch can show the complete cradle-to-completion lifecycle of every significant development in Ireland.

Estimated infrastructure cost increase: ~€5/month (Hetzner volume for expanded storage). Everything else fits on the existing CX32.

Section 14: The National Planning Application Database (NPAD)

14.1 What the NPAD Is

The Department of Housing, Local Government and Heritage operates a national ArcGIS Feature Service that aggregates planning application data from participating local authorities. It is published as open data under Creative Commons Attribution 4.0.

Live API endpoint:

```
https://services.arcgis.com/NzlPQPKn5QF9v2US/arcgis/rest/services/IrishPlanningApplications/FeatureServer/0
```

As of March 2026: - **362,721 planning applications** across **30 of 31 local authorities** - Updated **weekly** - Covers applications from **2010 onwards** - Full geometry (ITM coordinates) for mapping - **No authentication required**

14.2 Councils in the NPAD

All 30 councils confirmed present via live API query:

#	Council	#	Council
1	Carlow County Council	16	Limerick City & County Council
2	Cavan County Council	17	Longford County Council
3	Clare County Council	18	Louth County Council
4	Cork City Council	19	Mayo County Council
5	Donegal County Council	20	Meath County Council
6	**Dublin City Council** ✓ Phase 1	21	Monaghan County Council
7	Dún Laoghaire-Rathdown County Council	22	Offaly County Council
8	Fingal County Council	23	Roscommon County Council
9	Galway City Council	24	Sligo County Council
10	Galway County Council	25	South Dublin County Council
11	**Kerry County Council** ⚠ NOT in NPAD	26	Tipperary County Council
12	Kildare County Council	27	Waterford City & County Council
13	Kilkenny County Council	28	Westmeath County Council
14	Laois County Council	29	Wexford County Council
15	Leitrim County Council	30	Wicklow County Council
		31	South Dublin County Council

Kerry County Council is not in the NPAD and requires a separate integration (see Section 14.6).

14.3 NPAD Field Schema

The NPAD is extraordinarily rich. These fields are available for every application:

Field	Description	Phase 1 Equivalent
`PlanningAuthority`	Council name	*(new)*
`ApplicationNumber`	Planning reference	REG_REF
`DevelopmentDescription`	Proposal text	PROPOSAL
`DevelopmentAddress`	Site address	LOCATION
`DevelopmentPostcode`	Eircode	*(new)*
`ApplicantForename`	Applicant first name	*(scraped in Phase 1)*
`ApplicantSurname`	Applicant last name	*(scraped in Phase 1)*
`ApplicantAddress`	Applicant address	*(new)*
`ApplicationStatus`	Current status	STAGE
`ApplicationType`	Type code	APPTYPE
`Decision`	Grant/Refuse/etc.	DECISION
`LandUseCode`	Land use classification	*(new — very valuable)*
`AreaofSite`	Site area (numeric)	*(new)*
`NumResidentialUnits`	Number of units	*(new)*
`FloorArea`	Floor area m²	*(new)*
`OneOffHouse`	Boolean flag	*(new)*
`ReceivedDate`	Application submitted	APNDAT
`DecisionDate`	Decision made	DECDAT
`GrantDate`	Permission granted	FINAL_GRNT_DATE
`ExpiryDate`	Permission expiry	TIME_EXP
`AppealRefNumber`	ABP appeal reference	*(new)*
`AppealStatus`	Appeal status	*(new)*
`AppealDecision`	ABP decision	*(new)*
`AppealDecisionDate`	ABP decision date	*(new)*
`FIRequestDate`	Further info requested	*(new)*
`FIRecDate`	Further info received	*(new)*
`LinkAppDetails`	URL to council portal	*(new — direct link!)*
`ITMEasting` / `ITMNorthing`	Coordinates	spatial CSV

Critical observations: - `ApplicantForename` + `ApplicantSurname` are already in the NPAD — the Phase 1 portal scraper is no longer needed for national data - `FloorArea`, `NumResidentialUnits`, `AreaofSite` are numeric fields — the AI value estimation prompt can use these directly - `LandUseCode` gives a machine-readable classification baseline - `LinkAppDetails` provides direct deep links to each council's portal page - `OneOffHouse` boolean allows instant filtering of single-house extensions from commercially significant developments

14.4 NPAD Ingestion Architecture

The ArcGIS Feature Service paginates at 2,000 records per request. Full ingestion strategy:

```
# workers/npad_ingest.py

NPAD_URL = "https://services.arcgis.com/NzlPQPKn5QF9v2US/arcgis/rest/services/IrishPlanningApplications/FeatureServer/0/query"

async def ingest_npad_full():
    """Full ingestion: paginate through all 362k+ records."""
    offset = 0
    page_size = 2000
    total = await get_record_count()

    while offset < total:
        params = {
            "where": "1=1",
            "outFields": "*",
            "returnGeometry": "true",
            "outSR": "4326",          # Request WGS84 directly — no pyproj needed
            "f": "json",
            "resultOffset": offset,
            "resultRecordCount": page_size,
            "orderByFields": "OBJECTID ASC"
        }
        features = await fetch_page(NPAD_URL, params)
        upsert_applications_batch(features)
        offset += page_size

async def ingest_npad_incremental():
    """Incremental: only records updated since last sync.
    Use ETL_DATE field which tracks when each record was last touched."""
    last_sync = get_last_sync_timestamp()
    where = f"ETL_DATE > DATE '{last_sync.strftime('%Y-%m-%d')}'"
    # ... same pagination logic with where clause
```

Initial full load: ~182 paginated requests (362k ÷ 2000). At 1 req/sec, ~3 minutes. **Weekly incremental:** Only changed/new records since last ETL_DATE. Typically a few hundred records per day — seconds to sync.

14.5 Database Schema Extensions for National Data

Add one column to the `applications` table and one new table:

```
-- Add to applications table
ALTER TABLE applications ADD COLUMN planning_authority VARCHAR(100);
ALTER TABLE applications ADD COLUMN land_use_code VARCHAR(50);
ALTER TABLE applications ADD COLUMN area_of_site FLOAT;
ALTER TABLE applications ADD COLUMN num_residential_units INTEGER;
ALTER TABLE applications ADD COLUMN floor_area FLOAT;
ALTER TABLE applications ADD COLUMN one_off_house BOOLEAN;
ALTER TABLE applications ADD COLUMN applicant_forename VARCHAR(100);
ALTER TABLE applications ADD COLUMN applicant_surname VARCHAR(100);
ALTER TABLE applications ADD COLUMN applicant_address TEXT;
ALTER TABLE applications ADD COLUMN link_app_details TEXT;  -- direct portal URL
ALTER TABLE applications ADD COLUMN npad_object_id INTEGER; -- ArcGIS OBJECTID
ALTER TABLE applications ADD COLUMN data_source VARCHAR(20) DEFAULT 'dcc_csv';
-- data_source values: 'dcc_csv' | 'npad_arcgis' | 'kerry_apas' | 'manual'

-- Index on planning authority for national filtering
CREATE INDEX idx_applications_authority ON applications(planning_authority);
CREATE INDEX idx_applications_land_use ON applications(land_use_code);
CREATE INDEX idx_applications_units ON applications(num_residential_units);
CREATE INDEX idx_applications_floor_area ON applications(floor_area);

-- Composite index for national search (authority + date)
CREATE INDEX idx_applications_authority_date
    ON applications(planning_authority, apn_date DESC);
```

14.6 Kerry County Council — Separate Integration

Kerry is not in the NPAD. Kerry uses the same APAS system as DCC and publishes its own open data. Check:

```
https://data.gov.ie — search "Kerry planning"
https://www.kerrycoco.ie/planning/planning-applications/
```

Kerry's APAS data follows the same CSV format as DCC's open data. Once the URL is confirmed, add a `KerryIngestWorker` that mirrors the DCC ingest worker exactly — same field mapping, same upsert logic — with `data_source = 'kerry_apas'`.

If Kerry has no open data CSV, fall back to their planning portal search (same approach as the Phase 1 Agile portal scraper). Kerry applications represent ~1.5% of national volume — this is a low-priority integration.

Section 15: Post-Planning Lifecycle Tracking via BCMS

15.1 The BCMS Open Data Portal

The National Building Control and Market Surveillance Office (NBC&MSO) publishes building control data at **data.nbco.gov.ie** — free, no authentication, updated regularly.

Two datasets are directly relevant:

Dataset A: Commencement Notices + Certificates of Compliance on Completion

URL: `https://data.nbco.gov.ie/dataset/bcnccc` **Download:** `https://data.nbco.gov.ie/dataset/2704a333-874d-46f5-b3bc-3673766bf816/resource/0774e781-7af8-46da-b623-872e74cf541e/download/buildingscnscccs.csv`

Covers all 31 local authorities since 2014. A Commencement Notice (CN) must be filed before any significant building work begins. A Certificate of Compliance on Completion (CCC) is filed when the building is certified complete.

Key CN fields: | Field | Description | |---|---| | `CN_Planning_Permission_Number` | **The join key to planning applications** | | `CN_Commencement_Date` | Date construction started | | `CN_Date_Granted` | Date planning was granted | | `CN_Date_Expiry` | Date permission expires | | `CN_Description_proposed_development` | Project description | | `CN_Total_floor_area_of_building` | Floor area m² | | `CN_Total_Number_of_Dwelling_Units` | Total units | | `CN_Total_apartments` | Apartment count | | `CN_Number_stories_above_ground` | Number of floors | | `CN_Number_bedrooms` | Bedroom count | | `CN_Number_room` | Room count | | `CN_LAT` / `CN_LNG` | WGS84 coordinates | | `CN_Street`, `CN_Town`, `CN_Eircode`, `CN_County` | Full address | | `CN_Project_Status` | Current status | | `CN_Proposed_use_of_building_desc` | Proposed use description | | `CN_Protected_structure` | Protected structure flag | | `LocalAuthority` | Which council |

Key CCC fields (on same row as CN): | Field | Description | |---|---| | `CCC_Date_Validated` | Date completion certificate issued | | `CCC_Units_Completed` | Units actually completed | | `CCC_Type_of_Completion_Certificate` | Certificate type |

Dataset B: FSC / DAC Applications (Fire Safety + Disability Access)

URL: `https://data.nbco.gov.ie/dataset/applications` **Download:** `https://data.nbco.gov.ie/dataset/ae8f9134-fc5a-41e1-a150-bcccede234a0/resource/b1de41a3-e14c-40b7-b0d7-4e85d7a648fc/download/applications.csv`

Updated as of 17 March 2026. A Fire Safety Certificate application is required before construction of most commercial buildings and apartments. Filing an FSC is a strong signal that a developer is ready to build.

Key FSC fields: | Field | Description | |---|---| | `planning_permission_reference_no` | **Join key** | | `application_type` | FSC / DAC / Dispensation / Relaxation | | `local_authority` | Council | | `submission_date` | When FSC was applied for | | `date_of_decision` | FSC decision date | | `decision_type` | Granted / Refused | | `applicant_name` | Applicant (often company name) | | `floor_area_of_building` | Floor area m² | | `no_of_stories_above_ground_level` | Floors | | `total_combined_floor_area` | Combined floor area | | `use_of_proposed_works_or_building` | Use type | | `main_construction_type` | Construction type | | `site_area` | Site area | | `date_construction_started` | Construction start (if entered) | | `is_construction_of_building_complete` | Boolean | | `date_of_completion` | Completion date | | `LAT` / `LONGITUDE` | Coordinates | | `eircode` | Postcode |

15.2 The Complete Project Lifecycle

Joining all three data sources on planning reference number gives a complete lifecycle view:

```
STAGE 1: Application Submitted
  Source: NPAD / DCC CSV
  Signal: ReceivedDate populated
  Status: "Application Submitted"

STAGE 2: Registered
  Source: NPAD / DCC CSV
  Signal: RgnDate populated
  Status: "Registered"

STAGE 3: Further Information Requested (optional)
  Source: NPAD FIRequestDate / DCC FURINFO CSV
  Signal: FIRequestDate populated
  Status: "Further Information Requested"

STAGE 4: Decision Made
  Source: NPAD / DCC CSV
  Signal: DecisionDate + Decision populated
  Status: "Granted" | "Refused" | "Split Decision"

STAGE 5: Appeal Lodged (optional)
  Source: NPAD AppealStatus / DCC APPEAL CSV
  Signal: AppealRefNumber populated
  Status: "Appealed to An Bord Pleanála"

STAGE 6: Appeal Decided (optional)
  Source: NPAD AppealDecision
  Signal: AppealDecisionDate populated
  Status: "Appeal: Granted" | "Appeal: Refused"

STAGE 7: Fire Safety Certificate Applied For ← NEW (BCMS Dataset B)
  Source: BCMS FSC/DAC Applications
  Signal: submission_date populated, application_type = 'FSC'
  Status: "FSC Application Filed — Construction Imminent"
  Significance: Strong signal developer is ready to build

STAGE 8: Construction Started ← NEW (BCMS Dataset A)
  Source: BCMS Commencement Notice
  Signal: CN_Commencement_Date populated
  Status: "Under Construction"

STAGE 9: Building Complete ← NEW (BCMS Dataset A)
  Source: BCMS Certificate of Compliance on Completion
  Signal: CCC_Date_Validated populated
  Status: "Complete — Certificate of Compliance Issued"
```

This is a 9-stage lifecycle tracker, built entirely from open government data, covering every significant development in Ireland.

15.3 Database Schema: BCMS Tables

```
-- COMMENCEMENT NOTICES + COMPLETIONS
CREATE TABLE commencement_notices (
    id                          SERIAL PRIMARY KEY,
    reg_ref                     VARCHAR(30),  -- join key to applications.reg_ref
    local_authority             VARCHAR(100),

    -- Commencement
    cn_commencement_date        DATE,
    cn_proposed_end_date        DATE,
    cn_project_status           VARCHAR(50),
    cn_date_granted             DATE,
    cn_date_expiry              DATE,
    cn_description              TEXT,
    cn_proposed_use_desc        TEXT,

    -- Building characteristics (extremely useful for value estimation)
    cn_total_floor_area         FLOAT,
    cn_total_dwelling_units     INTEGER,
    cn_total_apartments         INTEGER,
    cn_number_stories_above     INTEGER,
    cn_number_bedrooms          INTEGER,
    cn_protected_structure      BOOLEAN,
    cn_phase                    VARCHAR(10),
    cn_units_for_phase          INTEGER,
    cn_total_phases             INTEGER,

    -- Address
    cn_street                   TEXT,
    cn_town                     TEXT,
    cn_eircode                  VARCHAR(10),
    cn_county                   VARCHAR(50),
    cn_lat                      FLOAT,
    cn_lng                      FLOAT,
    location_point              GEOMETRY(Point, 4326),

    -- Completion
    ccc_date_validated          DATE,
    ccc_units_completed         INTEGER,
    ccc_type                    VARCHAR(100),

    -- Metadata
    raw_data                    JSONB,
    ingested_at                 TIMESTAMPTZ DEFAULT NOW()
);

CREATE INDEX idx_cn_reg_ref ON commencement_notices(reg_ref);
CREATE INDEX idx_cn_date ON commencement_notices(cn_commencement_date DESC);
CREATE INDEX idx_cn_authority ON commencement_notices(local_authority);
CREATE INDEX idx_cn_location ON commencement_notices USING GIST(location_point);
CREATE INDEX idx_cn_ccc ON commencement_notices(ccc_date_validated)
    WHERE ccc_date_validated IS NOT NULL;

-- FIRE SAFETY / DISABILITY ACCESS CERTIFICATES
CREATE TABLE fsc_applications (
    id                              SERIAL PRIMARY KEY,
    reg_ref                         VARCHAR(30),  -- join key
    application_reference_no        VARCHAR(50),
    application_type                VARCHAR(20),  -- FSC | DAC | Dispensation | Relaxation
    local_authority                 VARCHAR(100),
    submission_date                 DATE,
    date_of_decision                DATE,
    decision_type                   VARCHAR(50),

    -- Building data
    floor_area_of_building          FLOAT,
    total_combined_floor_area       FLOAT,
    no_of_stories_above_ground      INTEGER,
    site_area                       FLOAT,
    use_of_proposed_works           TEXT,
    main_construction_type          VARCHAR(100),

    -- Construction status
    date_construction_started       DATE,
    is_construction_complete        BOOLEAN,
    date_of_completion              DATE,
    is_building_occupied            BOOLEAN,

    -- Applicant
    applicant_name                  TEXT,
    applicant_address_line_1        TEXT,
    applicant_town                  TEXT,
    applicant_county                VARCHAR(50),

    -- Location
    lat                             FLOAT,
    longitude                       FLOAT,
    eircode                         VARCHAR(10),
    location_point                  GEOMETRY(Point, 4326),
```

```
    raw_data                        JSONB,
    ingested_at                     TIMESTAMPTZ DEFAULT NOW()
);

CREATE INDEX idx_fsc_reg_ref ON fsc_applications(reg_ref);
CREATE INDEX idx_fsc_type ON fsc_applications(application_type);
CREATE INDEX idx_fsc_date ON fsc_applications(submission_date DESC);
CREATE INDEX idx_fsc_decision ON fsc_applications(decision_type);
```

15.4 Lifecycle Stage Computation

Add a computed `lifecycle_stage` field to the applications table, updated whenever related BCMS records are upserted:

```
ALTER TABLE applications ADD COLUMN lifecycle_stage VARCHAR(30);
ALTER TABLE applications ADD COLUMN lifecycle_updated_at TIMESTAMPTZ;

-- Values:
-- 'submitted'         — application received, not yet registered
-- 'registered'        — registered by planning authority
-- 'further_info'      — further information requested
-- 'decided_granted'   — permission granted
-- 'decided_refused'   — permission refused
-- 'appealed'          — under appeal to An Bord Pleanála
-- 'appeal_granted'    — ABP granted
-- 'appeal_refused'    — ABP refused
-- 'fsc_filed'         — Fire Safety Certificate applied for
-- 'under_construction'— Commencement Notice filed
-- 'complete'          — Certificate of Compliance on Completion issued
-- 'expired'           — permission expired without commencement
```

```
def compute_lifecycle_stage(app: Application) -> str:
    """Determine the current lifecycle stage in priority order."""
    # Completion takes highest priority
    if app.ccc_date_validated:
        return 'complete'
    if app.cn_commencement_date:
        return 'under_construction'
    if app.fsc_submission_date:
        return 'fsc_filed'
    # ABP appeal stages
    if app.appeal_decision_date:
        return 'appeal_granted' if 'grant' in app.appeal_decision.lower() else 'appeal_refused'
    if app.appeal_ref_number:
        return 'appealed'
    # Planning authority decision
    if app.decision_date:
        d = (app.decision or '').lower()
        if 'grant' in d:
            # Check if expired
            if app.expiry_date and app.expiry_date < date.today() and not app.cn_commencement_date:
                return 'expired'
            return 'decided_granted'
        return 'decided_refused'
    if app.fi_request_date:
        return 'further_info'
    if app.rgn_date:
        return 'registered'
    return 'submitted'
```

15.5 BCMS Ingest Workers

```
# workers/bcms_ingest.py

BCMS_CN_URL = "https://data.nbco.gov.ie/dataset/2704a333-874d-46f5-b3bc-3673766bf816/resource/0774e781-7af8-46da-b623-872e74cf541e/download/bu

BCMS_FSC_URL = "https://data.nbco.gov.ie/dataset/ae8f9134-fc5a-41e1-a150-bcccede234a0/resource/b1de41a3-e14c-40b7-b0d7-4e85d7a648fc/download/ap

async def ingest_bcms_commencements():
    """Download CN/CCC CSV and upsert all records."""
    df = await download_csv(BCMS_CN_URL)

    # Normalise planning reference number for joining
    # BCMS uses format like "23/1234" — NPAD uses "23/1234" or authority prefix
    df['reg_ref_normalised'] = df['CN_Planning_Permission_Number'].apply(normalise_ref)

    for row in df.itertuples():
        upsert_commencement_notice(row)
        # Update lifecycle stage on parent application
        update_lifecycle_stage(row.reg_ref_normalised)

async def ingest_bcms_fsc():
    """Download FSC/DAC CSV and upsert."""
    df = await download_csv(BCMS_FSC_URL)
    df['reg_ref_normalised'] = df['planning_permission_reference_no'].apply(normalise_ref)
    for row in df.itertuples():
        upsert_fsc_application(row)
        update_lifecycle_stage(row.reg_ref_normalised)

def normalise_ref(raw_ref: str) -> str:
    """Normalise planning references for cross-dataset joining.
    Handles variations like:
      'FRL/2023/12345' (DCC format)
      '23/1234'        (short county council format)
      'D23A/1234'      (Dublin with area code)
    Returns a canonical form for matching.
    """
    if not raw_ref:
        return None
    ref = raw_ref.strip().upper()
    # Remove spaces, normalise separators
    ref = re.sub(r'\s+', '', ref)
    return ref
```

Important note on join reliability: Planning reference number formats vary across councils. The `normalise_ref` function needs to handle multiple formats. Build a reference number normalisation table during initial ingestion to track match rates — aim for >90% match rate between BCMS and NPAD records. For unmatched BCMS records, store them separately and attempt fuzzy address-based matching as a fallback.

Section 16: AI Intelligence Layer

16.1 What This Replaces

CIS's human researchers do the following for each "significant" project:

1. Assign a construction value estimate (€)
2. Classify the development type in commercial terms
3. Decide whether the project is worth alerting subscribers about
4. Extract professional names (architect, planning consultant)
5. Write a short "project summary" for their database

Claude can do all five of these automatically and at scale, using data already in the database.

16.2 Construction Value Estimation

Irish Construction Cost Benchmarks (March 2026)

These are approximate benchmark figures based on published SCSI/Society of Chartered Surveyors Ireland data and industry norms:

Development Type	Unit	Cost Benchmark
Apartments (social/affordable)	per unit	€280,000 – €340,000
Apartments (private)	per unit	€320,000 – €420,000
Student accommodation	per bed	€90,000 – €130,000
Houses (one-off rural)	per house	€180,000 – €260,000
Houses (scheme)	per unit	€200,000 – €280,000
Hotel	per room	€180,000 – €280,000
Office	per m²	€2,800 – €4,200
Retail	per m²	€1,800 – €2,800
Industrial / warehouse	per m²	€800 – €1,400
Data centre	per m²	€8,000 – €15,000
Hospital / medical	per m²	€4,000 – €7,000
School / education	per m²	€2,200 – €3,500
Mixed use	blended calculation	—

Value Estimation Prompt

```
VALUE_ESTIMATION_PROMPT = """
You are estimating the construction value of an Irish planning application.

Use these Irish construction cost benchmarks (2026):
- Apartments (private): €350,000/unit average
- Apartments (social): €310,000/unit average
- Houses (scheme): €240,000/unit average
- One-off houses: €220,000/unit average
- Student accommodation: €110,000/bed
- Hotel: €230,000/room
- Office: €3,500/m²
- Retail/restaurant: €2,300/m²
- Industrial/warehouse: €1,100/m²
- Data centre: €11,000/m²
- Mixed use: calculate each component separately

Application data:
  Reference: {reg_ref}
  Authority: {planning_authority}
  Description: {description}
  Land Use Code: {land_use_code}
  Floor Area: {floor_area} m²
  Number of Residential Units: {num_units}
  Site Area: {site_area} m²
  Development Category: {dev_category}
  Stories Above Ground: {stories} (from BCMS if available)

Instructions:
1. Identify the primary development type
2. Select the most appropriate benchmark
3. Calculate a low and high estimate
4. State your primary basis (units × rate OR m² × rate)
5. Flag if data is insufficient for reliable estimate

Respond ONLY in valid JSON:
{{
  "value_low": 0,
  "value_high": 0,
  "primary_basis": "45 units × €320,000",
  "development_type_for_valuation": "Private Apartments",
  "confidence": "high|medium|low",
  "notes": "optional explanation"
}}
"""
```

Cost: Claude Haiku at ~$0.25/1M input tokens. Average prompt ~400 tokens. 362k applications × 400 tokens = ~145M tokens = **~$36 one-time backfill**. Ongoing: ~2,000 new applications/week nationally = <$1/week.

Estimated Value Storage

```
ALTER TABLE applications ADD COLUMN est_value_low     BIGINT;  -- in euros
ALTER TABLE applications ADD COLUMN est_value_high    BIGINT;
ALTER TABLE applications ADD COLUMN est_value_basis   TEXT;
ALTER TABLE applications ADD COLUMN est_value_type    VARCHAR(100);
ALTER TABLE applications ADD COLUMN est_value_confidence VARCHAR(10);
ALTER TABLE applications ADD COLUMN value_estimated_at TIMESTAMPTZ;

-- Index for filtering by value (commercial intelligence use case)
CREATE INDEX idx_applications_value ON applications(est_value_high DESC)
    WHERE est_value_high IS NOT NULL;
```

16.3 Significance Scoring

Not all applications are commercially interesting. CIS focuses on projects worth following. Automate this with a significance score:

```
def compute_significance_score(app: Application) -> int:
    """Score 0-100. >50 = significant. >80 = high priority."""
    score = 0

    # Value-based scoring (40 points max)
    value = app.est_value_high or 0
    if value >= 50_000_000:   score += 40
    elif value >= 10_000_000: score += 30
    elif value >= 2_000_000:  score += 20
    elif value >= 500_000:    score += 10

    # Unit count (20 points max)
    units = app.num_residential_units or 0
    if units >= 100:  score += 20
    elif units >= 50: score += 15
    elif units >= 20: score += 10
    elif units >= 5:  score += 5

    # Development type (20 points max)
    cat = app.dev_category or ''
    if cat in ('hotel_accommodation', 'commercial_office', 'industrial_warehouse'):
        score += 20
    elif cat in ('residential_new_build', 'mixed_use'):
        score += 15
    elif cat in ('commercial_retail',):
        score += 10

    # Decision outcome (10 points max)
    if app.decision and 'grant' in app.decision.lower():
        score += 10

    # Explicitly NOT one-off house (10 points)
    if not app.one_off_house:
        score += 10

    return min(score, 100)
```

```
ALTER TABLE applications ADD COLUMN significance_score INTEGER DEFAULT 0;
CREATE INDEX idx_applications_significance ON applications(significance_score DESC);
```

16.4 Professional Identification

The planning application form (available in documents for applications that have been scraped) always names: - The **planning agent/consultant** (the professional who submitted the application) - The **architect or design firm** (on all drawings)

For applications where documents have been indexed (Phase 1 Section 13, Tier C), Claude extracts these from the planning application form PDF using structured extraction:

```
PROFESSIONAL_EXTRACTION_PROMPT = """
Extract professional names from this Irish planning application form text.

Text:
{form_text}

Extract:
1. Planning Agent / Agent name and company
2. Architect / Designer name and company
3. Engineer name and company (if mentioned)

Respond ONLY in valid JSON:
{{
  "planning_agent_name": null,
  "planning_agent_company": null,
  "architect_name": null,
  "architect_company": null,
  "engineer_name": null,
  "engineer_company": null
}}
Return null for any field not found. Do not guess.
"""
```

```
ALTER TABLE applications ADD COLUMN planning_agent_name    TEXT;
ALTER TABLE applications ADD COLUMN planning_agent_company TEXT;
ALTER TABLE applications ADD COLUMN architect_name         TEXT;
ALTER TABLE applications ADD COLUMN architect_company      TEXT;
```

16.5 The Weekly Significant Approvals Digest

This is CIS's most-used product feature: a weekly alert of newly granted planning permissions above a threshold. Implement as an automated job and public RSS feed.

```
# workers/digest.py — runs every Monday at 6am

async def generate_weekly_digest():
    """Find all newly granted significant applications in the past 7 days."""
    new_grants = await db.fetch("""
        SELECT
            a.reg_ref,
            a.planning_authority,
            a.location,
            a.applicant_forename || ' ' || a.applicant_surname AS applicant,
            a.proposal,
            a.dev_category,
            a.dev_subcategory,
            a.num_residential_units,
            a.floor_area,
            a.est_value_low,
            a.est_value_high,
            a.est_value_basis,
            a.decision_date,
            a.link_app_details
        FROM applications a
        WHERE a.decision_date >= NOW() - INTERVAL '7 days'
          AND a.decision ILIKE '%grant%'
          AND a.significance_score >= 50
        ORDER BY a.est_value_high DESC NULLS LAST
    """)

    # Publish as:
    # 1. RSS feed at /feed/weekly-digest.xml
    # 2. JSON API at /api/digest/latest
    # 3. Optional: email to registered subscribers
    publish_digest(new_grants)
```

RSS feed structure — makes PlanSearch subscribable in any RSS reader:

```
<rss version="2.0">
  <channel>
    <title>PlanSearch — Weekly Significant Approvals</title>
    <link>https://plansearch.ie</link>
    <description>Newly granted planning permissions in Ireland this week</description>
    <item>
      <title>€42m Residential Development, Cherrywood, Dublin 18</title>
      <description>
        Grant of permission for 180 apartments across 3 blocks.
        Applicant: Cherrywood Developments DAC.
        Est. value: €42m–€56m. Reference: DLR/2024/12345
      </description>
      <link>https://plansearch.ie/application/DLR/2024/12345</link>
      <pubDate>Mon, 10 Mar 2026 06:00:00 +0000</pubDate>
    </item>
  </channel>
</rss>
```

Section 17: Updated System Architecture for Phase 2

17.1 New Data Flow

```
┌──────────────────────────────────────────────────────────────────────┐
│                        DATA SOURCES                                  │
│                                                                      │
│  DCC CSV (nightly)     NPAD ArcGIS (weekly)     Kerry APAS (weekly) │
│  data.smartdublin.ie   services.arcgis.com       kerrycoco.ie        │
│         │                      │                        │            │
│         └──────────────────────┴────────────────────────┘            │
│                                │                                     │
│                    BCMS CNs + CCCs (weekly)                          │
│                    data.nbco.gov.ie                                  │
│                                │                                     │
│                    BCMS FSC/DAC (weekly)                             │
│                    data.nbco.gov.ie                                  │
└──────────────────────────────┬───────────────────────────────────────┘
                               │
                     ┌─────────▼──────────┐
                     │  PostgreSQL + PostGIS│
                     │  applications       │
                     │  commencement_notices│
                     │  fsc_applications   │
                     │  companies (CRO)    │
                     │  application_docs   │
                     └────────┬───────────┘
                              │
              ┌───────────────┴────────────────┐
              │                                │
    ┌─────────▼─────────┐  ┌─────────▼─────────┐  ┌─────────▼─────────┐
    │ Claude API        │ │ CRO API           │ │ Document          │
    │ Classification    │ │ Enrichment        │ │ Scraper           │
    │ Value Est.        │ │                   │ │ (Section 13)      │
    │ Significance      │ └───────────────────┘ └───────────────────┘
    │ Professional ID   │
    └───────────────────┘
                              │
                     ┌─────────▼──────────┐
                     │  FastAPI Backend    │
                     │  + Redis Workers    │
                     └────────┬───────────┘
                              │
                     ┌─────────▼──────────┐
                     │  Next.js Frontend   │
                     │  National map       │
                     │  Lifecycle tracker  │
                     │  Digest / RSS       │
                     │  Admin centre       │
                     └────────────────────┘
```

17.2 Updated Database Size Estimates

Table	Phase 1 (DCC only)	Phase 2 (National)
applications	~200k rows, ~800MB	~650k rows, ~3GB
commencement_notices	—	~200k rows, ~1.5GB
fsc_applications	—	~80k rows, ~600MB
application_documents (metadata)	~3M rows	~12M rows
companies	~20k rows	~80k rows
Total	**~1.5GB**	**~8GB**

The Hetzner CX32's 80GB NVMe drive handles this comfortably. Add a 20GB Hetzner Volume (€0.95/month) for document PDF archive if implementing Tier C document indexing.

17.3 Cron Schedule Updates

```
# /etc/cron.d/plansearch-phase2

# Nightly DCC CSV sync (unchanged from Phase 1)
0 2 * * *   python -m app.workers.dcc_ingest

# Weekly NPAD sync (Sunday night)
0 1 * * 0   python -m app.workers.npad_ingest --mode incremental

# Weekly BCMS sync (Sunday night, after NPAD)
30 1 * * 0  python -m app.workers.bcms_ingest --dataset commencements
45 1 * * 0  python -m app.workers.bcms_ingest --dataset fsc

# Weekly lifecycle stage recomputation (Sunday night, after BCMS)
0 3 * * 0   python -m app.workers.lifecycle_updater

# AI enrichment: value estimation + significance scoring (Monday early morning)
0 4 * * 1   python -m app.workers.ai_enrichment --batch-size 500

# Weekly digest generation (Monday 6am)
0 6 * * 1   python -m app.workers.digest_generator

# Existing Phase 1 jobs continue unchanged
0 3 * * *   python -m app.workers.classifier --batch-size 200
0 1 * * *   python -m app.workers.scraper --hours 5  # legacy DCC portal
```

Section 18: Phase 2 Frontend — National Search UI

18.1 New UI Elements

National Authority Filter

Add a council selector to the filter panel. Group by province for usability:

```
[All Ireland ▾]
  Leinster
    ☐ Dublin City Council
    ☐ Dún Laoghaire-Rathdown
    ☐ Fingal
    ☐ South Dublin
    ☐ Kildare, Meath, Wicklow, Louth, Wexford, Carlow, Kilkenny, Laois, Offaly, Longford, Westmeath
  Munster
    ☐ Cork City, Cork County, Kerry, Limerick, Tipperary, Waterford, Clare
  Connacht
    ☐ Galway City, Galway County, Mayo, Roscommon, Sligo, Leitrim
  Ulster (ROI)
    ☐ Donegal, Cavan, Monaghan
```

Lifecycle Stage Filter

```
[Project Stage ▾]
  ☐ Application Submitted
  ☐ Registered
  ☐ Pending Decision
  ☐ Granted
  ☐ Refused
  ☐ Under Appeal
  ☐ FSC Filed — Construction Imminent  ← new
  ☐ Under Construction                 ← new
  ☐ Complete                           ← new
  ☐ Permission Expired                 ← new
```

Value Range Filter (for significant projects)

```
[Est. Value ▾]
  ☐ Any
  ☐ €500k+
  ☐ €2m+
  ☐ €10m+
  ☐ €50m+
  ☐ €100m+
```

National Map View

The Phase 1 map (Dublin only) becomes a national map with: - All-Ireland base layer (OpenStreetMap) - Province/county boundary overlays (from CSO open data) - Cluster markers that resolve to individual pins on zoom - Heat map toggle showing application density by area - Filter by lifecycle stage changes pin colour: - Grey = submitted/registered - Amber = pending decision - Green = granted - Red = refused - Blue = under construction - Teal = complete

18.2 Application Detail Page Enhancements

Add the lifecycle timeline panel to every application page:

```
┌──────────────────────────────────────────────────────────┐
│  PROJECT LIFECYCLE                                        │
├──────────────────────────────────────────────────────────┤
│  ✅ Submitted      12 Jan 2023                            │
│  ✅ Registered     19 Jan 2023                            │
│  ✅ Decision Made  28 Mar 2023  →  GRANTED                │
│  ✅ FSC Filed      14 Jun 2023  →  Granted 02 Aug 2023   │
│  🔨 Construction Started  03 Sep 2023                     │
│  ⏳ Completion Certificate  —  Pending                    │
│                                                           │
│  Est. Construction Value: €28m – €36m                     │
│  Basis: 95 units × €320,000 avg                           │
└──────────────────────────────────────────────────────────┘
```

18.3 New Public Pages

/significant — The "CIS Killer" Page

A public, filterable feed of the most commercially significant planning activity in Ireland this week. Default view: granted permissions, €2m+, sorted by estimated value descending. Updated Monday mornings.

```
🏗 Significant Planning Activity — Week of 10 March 2026

[Filter: All Ireland ▾] [€2m+ ▾] [Granted ▾] [All Types ▾]

1. €125m   An Droichead Órga, Mixed Use, Inchicore, Dublin
           DCC | Granted 09 Feb 2026 | 320 units + retail
           Applicant: Inchicore Regeneration DAC

2. €65m    LRD Residential Development, Blessington, Co. Wicklow
           Wicklow CC | Granted 09 Feb 2026 | 210 units
           Applicant: Glenmore Homes Ltd

3. €48m    Cherrywood Residential Development, Dublin 18
           DLR | Updated 05 Mar 2026 | 140 units
           ...
```

/feed/weekly-digest.xml — Public RSS Feed

Subscribable by anyone in any RSS reader or client. Updated weekly.

/api/digest/latest — JSON API

Free, open, no auth required. Returns current week's significant approvals as JSON. Allows third-party apps to build on PlanSearch data.

Section 19: What PlanSearch Phase 2 Becomes vs. CIS

19.1 Feature Comparison

Feature	CIS (paid, ~€3,000/yr)	PlanSearch Phase 2 (free)
National coverage	✅ All Ireland	✅ All 31 councils
Planning application data	✅	✅
Applicant name	✅	✅ (from NPAD)
Application documents	❌ (links only)	✅ (full index + text)
Construction value estimate	✅ (human research)	✅ (AI estimated)
Project stage tracking	✅ (human research)	✅ (automated via BCMS)
Construction start date	✅	✅ (Commencement Notice)
Completion date	✅	✅ (CCC)
Architect / agent identification	✅ (human)	✅ (AI extraction from docs)
FSC / DAC status	❌	✅
Weekly significant approvals digest	✅ (email)	✅ (RSS + JSON API)
CRO company enrichment	Partial	✅ (full API)
Map view	✅	✅
Full-text document search	❌	✅ (Phase 2 Tier C)
Free public access	❌	✅
Open API	❌	✅
Update frequency	Weekly (human)	Weekly (automated)

19.2 What CIS Still Has That PlanSearch Doesn't

Two genuine gaps remain after Phase 2:

1. Pre-planning intelligence. CIS has industry contacts who tip them off about projects before they hit the planning system — large LADs (Letters of Applicant's Determination), pre-planning meeting records, developer announcements. This requires a human network and cannot be automated from open data.

2. Contractor / tender award tracking. Once a project is under construction, CIS tracks who won the main contract and key subcontracts. eTenders.gov.ie covers public contracts but private construction tendering is not publicly reported. This remains a gap.

Everything else CIS does can now be replicated or exceeded by PlanSearch at zero cost to users.

Section 20: Phase 2 Implementation Plan

Timeline

Month 1: National Data Foundation

- [ ] Extend database schema with all new tables and columns (Section 14.5, 15.3, 16)
- [ ] Build and test NPAD ArcGIS ingest worker (Section 14.4)
- [ ] Run initial full NPAD ingest (362k records, ~3 minutes)
- [ ] Build and test BCMS CN/CCC ingest worker (Section 15.5)
- [ ] Build and test BCMS FSC/DAC ingest worker (Section 15.5)
- [ ] Implement `normalise_ref` join logic, measure match rates
- [ ] Implement `compute_lifecycle_stage` function
- [ ] Run lifecycle recomputation across all records
- [ ] Verify data integrity: sample 100 applications, check all fields

Month 2: AI Enrichment

- [ ] Build value estimation worker (Section 16.2)
- [ ] Store cost benchmarks as admin-configurable table (updatable without code deploy)
- [ ] Run value estimation backfill — all applications with sufficient data
- [ ] Build significance scoring function (Section 16.3)
- [ ] Build professional identification worker (Section 16.4) — for Doc Tier C records
- [ ] Add all new fields to search API filters
- [ ] Add est_value and significance_score to search results

Month 3: Frontend National Expansion

- [ ] National authority filter with province grouping (Section 18.1)
- [ ] Lifecycle stage filter
- [ ] Value range filter
- [ ] National map (Ireland-wide, clustered, colour by lifecycle stage)
- [ ] Application detail page: lifecycle timeline panel
- [ ] /significant page — weekly digest feed UI
- [ ] RSS feed endpoint
- [ ] Public JSON API endpoint

Month 4: Kerry + Digest + Polish

- [ ] Research Kerry County Council data availability and integrate (Section 14.6)
- [ ] Automated weekly digest job (Section 16.5)
- [ ] Email digest subscription (optional — simple signup, no auth required)
- [ ] Performance tuning for 650k+ records
- [ ] Load testing at national scale
- [ ] Mobile polish for national map view

Section 21: Phase 2 Cost Summary

Item	Monthly Cost
Hetzner CX32 (unchanged)	€8.19
Hetzner Volume 20GB (document storage)	€0.95
Claude API — value estimation backfill (one-time)	~€34
Claude API — ongoing weekly enrichment	~€2
Domain + SSL (unchanged)	~€1
Total ongoing	**~€12/month**

Phase 2 remains under €15/month. The entire national planning intelligence platform for Ireland — planning applications, lifecycle tracking, AI value estimates, company enrichment, document indexing, and weekly digest — costs less per month than a single CIS user pays per day.

Section 22: Notes for Claude Code — Phase 2 Build

1. **Start with the NPAD ingest.** Run it, verify record counts by council, check field quality on 100 sample records. Fix field mapping issues before building anything else.
2. **The planning reference normalisation problem is the hardest part.** Councils use different reference formats. Build `normalise_ref` incrementally, logging every format variation encountered. Aim for >90% automatic match rate between NPAD and BCMS. Accept that ~10% will be unmatched and handle gracefully.
3. **NPAD data quality varies by council.** Some councils populate `FloorArea` and `NumResidentialUnits` consistently; others do not. Build the AI value estimator to work with partial data and express lower confidence when fields are missing.
4. **Run the value estimator only on applications with sufficient data.** Minimum requirement: either (FloorArea > 0) OR (NumResidentialUnits > 0) OR (description length > 50 chars). Anything below this threshold is likely a trivial application (fence, gate, shed) — skip it.
5. **The significance score is your quality filter.** Set the threshold for the `/significant` page at 50 by default, adjustable in admin config. Check that the weekly output looks sensible — roughly 50–150 projects per week nationally at score ≥ 50.
6. **The national map needs clustering.** 650k pins without clustering will crash any browser. Use `supercluster` (React) or PostGIS `ST_ClusterDBSCAN` for server-side clustering at lower zoom levels. Only load individual pins when zoomed to county level or closer.
7. **Keep the Phase 1 DCC pipeline running unchanged.** The NPAD and DCC CSV cover the same records but have different update frequencies (weekly vs nightly) and slight field differences. Continue running the DCC CSV nightly sync for Dublin — it ensures fresher data for the biggest council. Use `data_source` column to track origin; where records overlap, prefer the most recently updated.
8. **The RSS feed is a growth driver.** Make it excellent. Include estimated value, development type, applicant, and a clean description in every item. This is the feature that will get PlanSearch shared by architects, contractors, and property journalists.

End of PlanSearch Phase 2 Technical Specification v1.0 Combine with PlanSearch Phase 1 Specification v1.1 for complete build reference. Start each Claude Code session with: `claude --dangerously-skip-permissions`